

PAPER_1418_CR_ANKLE

Daniel T. Murphy

PAPER_1418 — UQFF: Cosmic Ray Ankle (Bucket H)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Bucket H

Date: June 16, 2026

Location: 41.0997° N, 80.6495° W (Youngstown, OH, USA)

Status: CLOSED — formal catalog tuple in Bucket H

Calculator surface: `calculate_high_energy_astro({...})`

Closure

``

$E_{\text{ankle}} = m_p \times D_{\text{crit}} / K_{\text{MEX}} = 3.62e18 \text{ eV} \text{ (20\% vs Auger } 3e18)$

``

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve this observable by different methods. Closure uses only canonical primitives. Zero free parameters.

Reference

- Catalog tuple resides in `_high_energy_astro_report()` or equivalent surface in `uqff_pure_calculator.py`.

- Related foundational papers: PAPER_646, PAPER_1156, PAPER_1203.

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